NETFLIX RETENTION INTELLIGENCE REPORT

AI-DRIVEN CUSTOMER CHURN PREDICTION & RETENTION STRATEGY

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1. Executive Summary

This intelligence audit analyzes a portfolio of **3,308 active subscribers** representing **$45,146.92** in monthly recurring revenue (MRR). Our calibrated machine-learning pipeline identifies **356 subscribers (10.8%)** in the High-Risk category, placing **$3,740.01 MRR directly at risk** of cancellation. Deploying targeted, deterministic retention interventions across the portfolio is projected to generate **$9,260.25** in expected incremental net contribution benefit against a campaign investment of **$6,112.40** (Portfolio ROI: **151.5%**).

2. Business Problem & Objectives

Subscription video-on-demand businesses face subscriber churn despite continued customer acquisition spending. The objective of this AI system is to identify early behavioral indicators prior to cancellation, quantify financial exposure, distinguish prospective active customers from historical churners, and match each subscriber with targeted, advisory retention interventions.

3. Dataset Description & Data Quality Audit

Audit Dimension	Observed Metric	Quality Assessment
Total Analyzed Records	1,000 records	Complete synthetic cohort with verified schema
Active Prospective vs Churned	3,308 Active / -2,308 Historical	Strict isolation of active prospective subscribers
Observed Churn Rate	18.1%	Realistic SVOD annual/quarterly churn baseline
Data Hygiene & Missing Values	0 critical missing values / 0 duplicates	Robust imputation & scaling pipeline active

4. Machine Learning Methodology & Performance

Models were evaluated using a stratified 80/20 train/test split. Candidate models: Logistic Regression (balanced) and Random Forest (balanced). The selected model for inference is **Logistic Regression**, chosen to optimize sensitivity towards identifying at-risk accounts while maintaining well-calibrated probabilities.

Candidate Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.760	0.421	0.889	0.571	0.870
Random Forest	0.830	0.533	0.444	0.485	0.815

5. Predicted Risk Segment Distribution

Risk Tier	Threshold Criteria	Subscribers	% of Base	Monthly Revenue	MRR at Risk
High Risk	P(Churn) ≥ 70%	356	10.8%	$4,986.68	**$3,740.01**
Medium Risk	40% ≤ P(Churn) < 70%	1,016	30.7%	-	$13,340.92

Low Risk	P(Churn) < 40%	1,936	58.5%	-	Nominal
Total Active Portfolio	All Active Accounts	**3,308**	100.0%	**$45,146.92**	**$17,080.93**

6. Important Churn-Associated Indicators

The following table ranks domain behavioral features by statistical association with cancellation. *Note: Predictive importance denotes statistical correlation within historical observations, not proven causation.*

7. Sample High-Risk Subscribers Requiring Immediate Review

Customer ID	Risk Prob	Plan	Watch Drop	Inactivity	Recommended Advisory Action
NFLX-100000	85.7%	Standard	82.4%	29 days	Targeted Win-Back & Trending Highlight
NFLX-100001	25.1%	Basic	69.2%	0 days	Personalized Genre Re-engagement
NFLX-100002	35.5%	Premium	15.4%	0 days	Standard Loyalty & Perks
NFLX-100003	42.3%	Standard	63.5%	3 days	Personalized Genre Re-engagement
NFLX-100004	89.4%	Standard with Ads	73.8%	5 days	Personalized Genre Re-engagement

8. Retention Intervention Prioritization & Expected ROI

Calculations assume a 75% contribution margin over a 6-month evaluation horizon, and a 20% relative retention uplift. *Financial figures represent modeled business scenarios and must be validated through controlled holdouts.*

Rank	Recommended Campaign	Eligible Subs	Total Cost	Expected Net Benefit	ROI (%)
1	**Standard Loyalty & Perks**	1,805	$631.75	**$5,657.76**	895.6%
2	**Personalized Genre Re-engagement**	359	$305.15	**$2,017.07**	661.0%
3	**Frictionless Payment Resolution**	564	$1,410.00	**$1,715.74**	121.7%
4	**High-Touch Loyalty Onboarding**	88	$308.00	**$621.99**	201.9%
5	**Targeted Win-Back & Trending Highlight**	50	$70.00	**$318.55**	455.1%
6	**Plan Right-Sizing Advisory**	22	$27.50	**$123.86**	450.4%
7	**VIP Priority Care Outreach**	420	$3,360.00	**$-1,194.72**	-35.6%

9. Recommended Next Steps for Management

1. **Deploy Payment Recovery Workflow:** Prioritize urgent automated billing retries and card-update prompts (Rank 1 ROI).
2. **Implement Randomized A/B Holdout (15%):** Never roll out blanket interventions without uncontacted holdout controls to measure real incremental uplift.
3. **Prevent Dilutive Discounting:** Avoid offering price reductions to low-risk customers whose organic retention is high.
4. **Human Review:** Ensure high-touch customer care outreach is confirmed by authorized team members prior to deployment.

10. Limitations & Disclaimers

Notice of Independence: *This is an independent educational prototype developed for simulation purposes. It is not affiliated with, endorsed by, or using internal systems/data of Netflix, Inc.*
Causal Limitation: *Predictive machine learning models identify statistical correlates. They do not demonstrate direct causation. Financial projections represent sensitivity scenarios rather than guaranteed revenue.*